CAPSTONE SYSTEMS SYNTHESIS

Module map

- Module 16: Capstone Systems Synthesis
- Connected lab: lab-17_capstone-systems-synthesis.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 16
Capstone Systems Synthesis

1. Biology across scales
2. Mechanism chains
3. Feedback and constraint
4. Evidence from multiple modules
5. Capstone explanation and revision

Lab evidence
lab-17_capstone-systems-synthesis.md

Teaching note: Open by naming the module claim and pointing students to the lab as the evidence surface.

CAPSTONE SYSTEMS SYNTHESIS

Learning objectives

- Connect concepts from at least three modules in one explanation.
- Trace a mechanism across biological scales.
- Identify feedback loops, constraints, and tradeoffs.
- Match evidence types to claims at different scales.
- Revise a systems explanation using new evidence.

OBJECTIVE 1
Connect concepts from at least three modules in one explanation.

OBJECTIVE 2
Trace a mechanism across biological scales.

OBJECTIVE 3
Identify feedback loops, constraints, and tradeoffs.

OBJECTIVE 4
Match evidence types to claims at different scales.

OBJECTIVE 5
Revise a systems explanation using new evidence.

Teaching note: Frame objectives as actions students should be able to perform, not facts to copy.

CAPSTONE SYSTEMS SYNTHESIS

Topic sequence

- Biology across scales: Connect molecules, cells, organisms, populations, and ecosystems through specific mechanisms, such as a plant variant affecting water use.
- Mechanism chains: Explain each causal link. Plant reproduction can alter allele frequencies; plant abundance can alter herbivore food. These require different evidence.
- Feedback and constraint: Balancing feedback counteracts change; reinforcing feedback amplifies it. Constraints limit outcomes; tradeoffs pair a benefit with a cost.
- Evidence from multiple modules: Match evidence to scale: sequences test variants, measurements test water use, seed counts test reproduction, and repeated counts test abundance.
- Capstone explanation and revision: Connect at least three course modules. State an alternative explanation and a distinguishing observation; revise the weakest link when evidence conflicts.

01

Biology across scales

Connect molecules, cells, organisms, populations, and ecosystems through specific mechanisms, such as a plant variant affecting water use.

02

Mechanism chains

Explain each causal link. Plant reproduction can alter allele frequencies; plant abundance can alter herbivore food. These require different evidence.

03

Feedback and constraint

Balancing feedback counteracts change; reinforcing feedback amplifies it. Constraints limit outcomes; tradeoffs pair a benefit with a cost.

04

Evidence from multiple modules

Match evidence to scale: sequences test variants, measurements test water use, seed counts test reproduction, and repeated counts test abundance.

05

Capstone explanation and revision

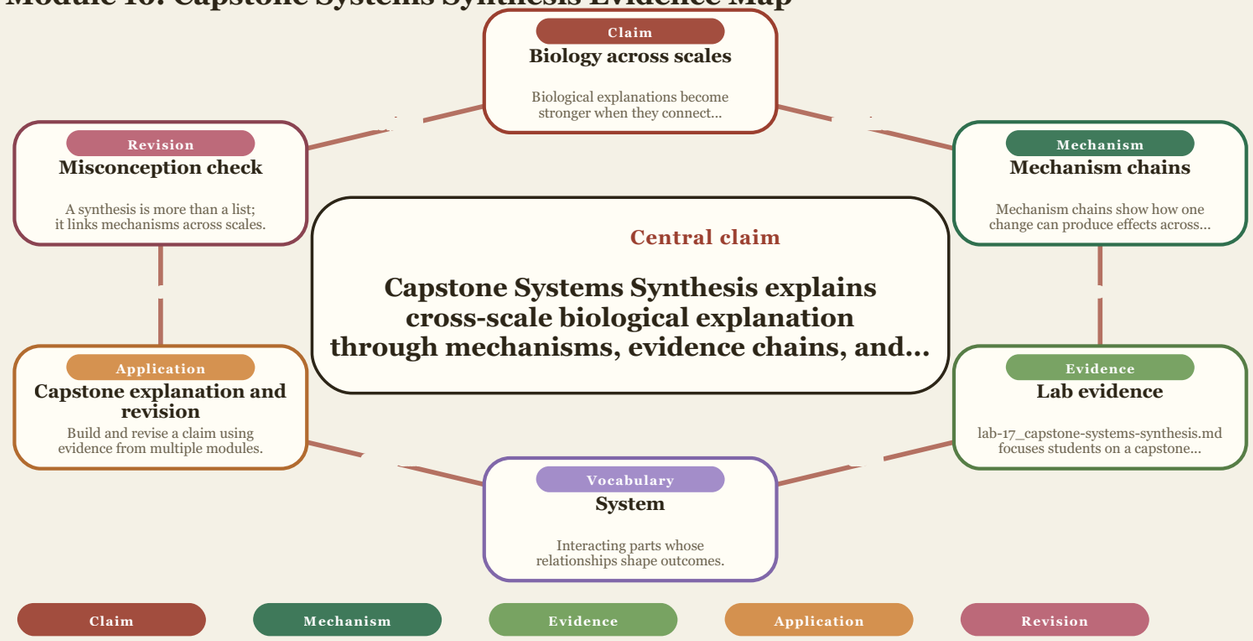
Connect at least three course modules. State an alternative explanation and a distinguishing observation; revise the weakest link when evidence conflicts.

Teaching note: Show the module as an ordered explanation so students can locate each new idea.

CAPSTONE SYSTEMS SYNTHESIS

Module 16: Capstone Systems Synthesis Evidence Map

- Module focus: Capstone Systems Synthesis
- Central claim: Capstone Systems Synthesis explains cross-scale biological explanation through mechanisms, evidence chains, and revision.
- Key nodes to connect: Biology across scales, Mechanism chains, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-17_capstone-systems-synthesis.md supplies an evidence surface for the map.

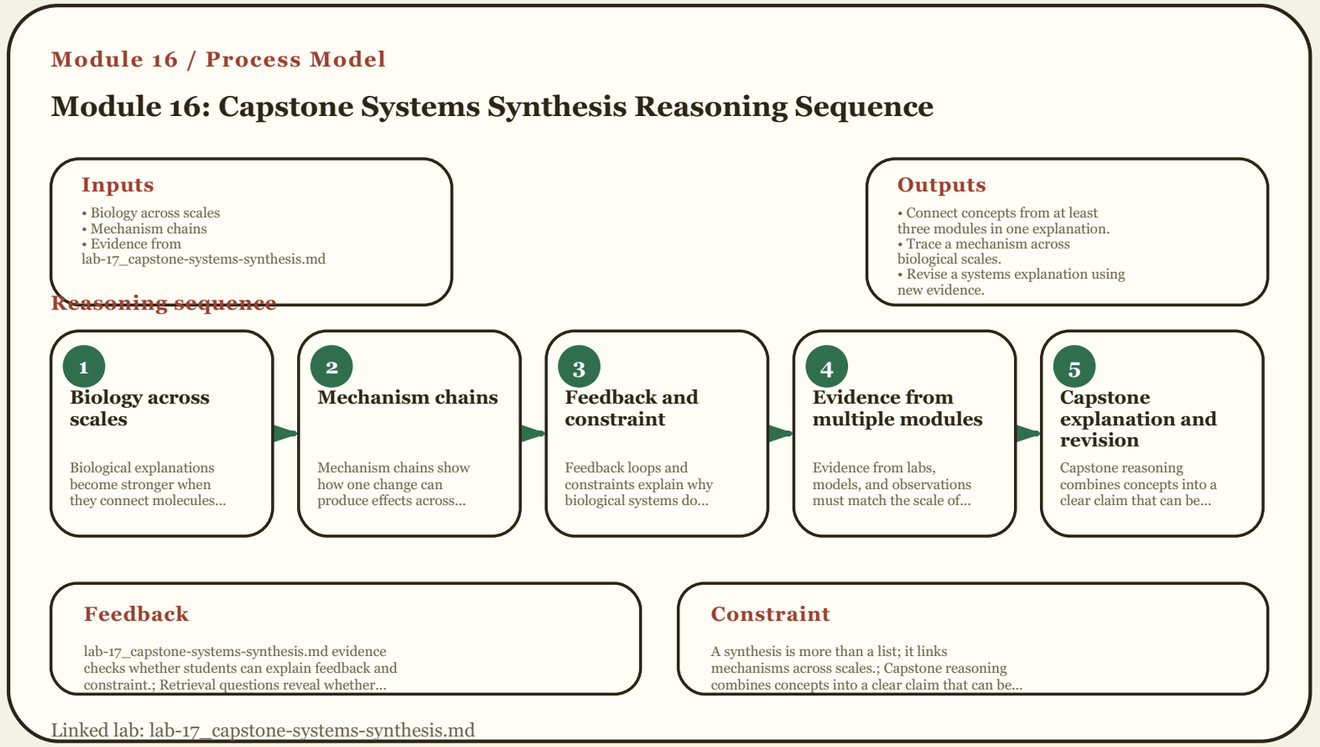


Teaching note: Ask students to explain one edge in their own words before moving on.

CAPSTONE SYSTEMS SYNTHESIS

Module 16: Capstone Systems Synthesis Reasoning Sequence

- Module focus: Capstone Systems Synthesis
- Trace the model: Biology across scales -> Mechanism chains -> Feedback and constraint -> Evidence from multiple modules
- Inputs become outputs: Biology across scales, Mechanism chains -> Connect concepts from at least three modules in one explanation., Trace a mechanism across biological scales.
- Feedback check: lab-17_capstone-systems-synthesis.md evidence checks whether students can explain feedback and constraint.
- Lab connection: lab-17_capstone-systems-synthesis.md gives students a process to observe or test.



Teaching note: Emphasize where the process can fail, slow down, or be revised by evidence.

CAPSTONE SYSTEMS SYNTHESIS

Terms as evidence handles

- System: Interacting parts whose relationships shape outcomes.
- Scale: Level of biological organization.
- Mechanism: Causal process that explains how something happens.
- Feedback loop: Circular causal relationship that amplifies or reduces change.
- Constraint: Limit that shapes what a system can do.
- Tradeoff: Benefit in one function paired with cost elsewhere.

SYSTEM
Interacting parts whose relationships shape outcomes.

SCALE
Level of biological organization.

MECHANISM
Causal process that explains how something happens.

FEEDBACK LOOP
Circular causal relationship that amplifies or reduces change.

CONSTRAINT
Limit that shapes what a system can do.

TRADEOFF
Benefit in one function paired with cost elsewhere.

Teaching note: Treat terms as handles for reasoning. Each term should help explain a claim or observation.

CAPSTONE SYSTEMS SYNTHESIS

Lab connection

- Lab file: lab-17_capstone-systems-synthesis.md
- Lab evidence should test or illustrate: Feedback and constraint
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Biology across scales

EVIDENCE
lab-17_capstone-systems-synthesis.md

CLAIM
Connect concepts from at least three modules in one explanation.

Teaching note: Make the lab connection explicit before students encounter the activity.

CAPSTONE SYSTEMS SYNTHESIS

Contrast check

- Do not stop at: Connect molecules, cells, organisms, populations, and ecosystems through specific mechanisms, such as a plant variant affecting water use.
- Stronger explanation: Connect at least three course modules. State an alternative explanation and a distinguishing observation; revise the weakest link when evidence conflicts.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Connect molecules, cells, organisms, populations, and ecosystems through specific mechanisms, such as a plant variant affecting water use.

DEEPER

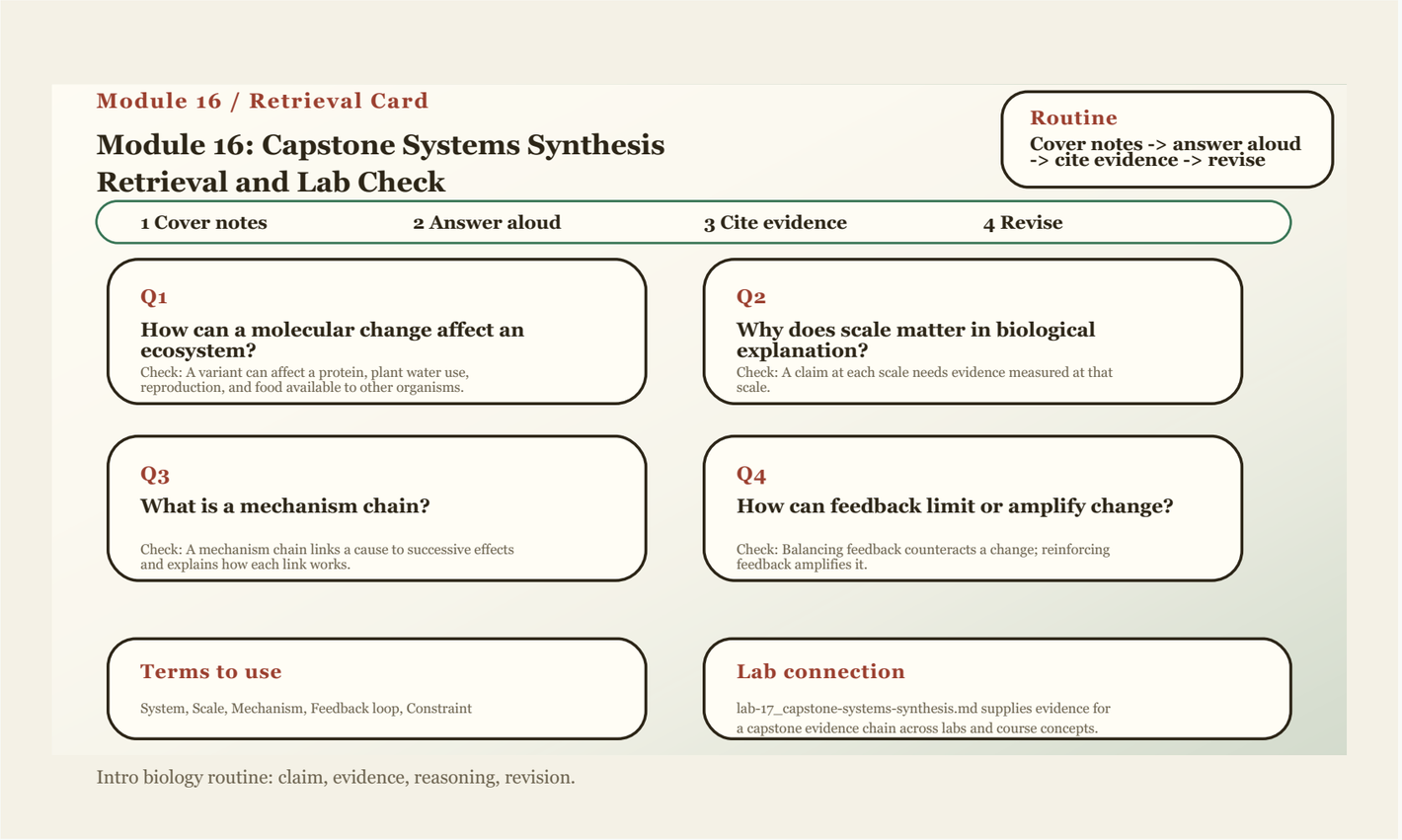
Connect at least three course modules. State an alternative explanation and a distinguishing observation; revise the weakest link when evidence conflicts.

Teaching note: Use this slide to prevent shallow vocabulary-only learning.

CAPSTONE SYSTEMS SYNTHESIS

Module 16: Capstone Systems Synthesis Retrieval and Lab Check

- Module focus: Capstone Systems Synthesis
- Retrieval prompt: How can a molecular change affect an ecosystem?
- Answer check: A variant can affect a protein, plant water use, reproduction, and food available to other organisms.
- Required terms: System, Scale, Mechanism
- Lab connection: lab-17_capstone-systems-synthesis.md; lab-17_capstone-systems-synthesis.md supplies evidence for a capstone evidence chain across labs and course concepts.



Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

CAPSTONE SYSTEMS SYNTHESIS

Practice quiz bridge

- 1. A strong biological explanation that spans scales does what?
- 2. When rising body temperature makes you sweat and cool back down, that is an example of:
- 3. "A larger body helps an animal retain heat but also requires more food." This statement describes a:
- 4. To support a claim that spans several biological scales, the strongest evidence is:

QUESTION 1

Key: C

A strong biological explanation that spans scales does what?

QUESTION 2

Key: A

When rising body temperature makes you sweat and cool back down, that is an example of:

QUESTION 3

Key: D

"A larger body helps an animal retain heat but also requires more food." This statement describes a:

QUESTION 4

Key: B

To support a claim that spans several biological scales, the strongest evidence is:

Teaching note: Use the quiz as formative feedback. Answer key: 1=C, 2=A, 3=D, 4=B

CAPSTONE SYSTEMS SYNTHESIS

Synthesis and exit ticket

- Exit claim: explain biology across scales using evidence from lab-17_capstone-systems-synthesis.md.
- Must include: System, Scale, Mechanism.
- Revision prompt: what evidence would change your explanation?

CLAIM
Biology across scales

EVIDENCE
lab-17_capstone-systems-synthesis.md

REVISION
What would change your mind?

Teaching note: End with a claim-evidence-reasoning exit ticket connected to the same module data.